

# SAI TEJA SRIVILLIBHUTTURU

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## Skills

**ML Models & Algorithms:** Python, PyTorch, XGBoost, LightGBM, CTR/CVR prediction, Learning-to-Rank (TF-Ranking), GBDT, deep learning (DNN, embedding layers)

**Recommendation Systems:** Collaborative filtering (SVD), large-scale ranking, real-time inference (<1ms latency), feature engineering, PySpark pipelines

**Production ML:** Real-time bidding systems, auction mechanisms, budget pacing, A/B testing, distributed systems (Kafka, Redis), AWS (S3, Lambda, SageMaker), Kubernetes

## Professional Experience

### University of Texas at Arlington

Aug 2024 – Present

*Machine Learning Engineer*

Texas, US

- Engineered **large-scale ML** system for information retrieval over 1K+ document corpus, achieving **92% accuracy** with **sub-100ms** query latency on AWS; communicated architecture decisions with 5-person team and presented optimization results to stakeholders.
- Constructed **real-time ML data pipeline** in PyTorch on distributed compute; trained and optimized path prediction models, cutting end-to-end routing latency **35%**; collaborated with infrastructure team to resolve compute bottlenecks.
- Formulated high-throughput **model inference** serving layer with RESTful APIs in Python; scaled to 50K+ daily requests and reduced serving latency **25%**.

### DentalScan

Dec 2025 – Feb 2026

*Machine Learning Engineer Intern*

Remote, US

- Automated ML model retraining on AWS + Docker + Kubernetes with Python CI/CD pipelines; ingested label-correction events via Kafka to trigger continuous model retraining.
- Reduced ML model retraining cycle **50%** and extended inference endpoints to **sub-200ms p99 latency**; improved model throughput via PyTorch-optimized serving on AWS.

### Tata Consultancy Services

Jun 2019 – May 2023

*Software Development Engineer – Backend & Infrastructure*

Chennai, India

- Led and scaled microservices backend exposing high-throughput RESTful APIs processing millions of records; slashed latency **45%** via MySQL optimization and Redis caching; coordinated with 3 teams on API contract changes.
- Established real-time monitoring service tracking microservice health; cut incident response time **40%** via Prometheus/Grafana dashboards; liaised with oncall team to gather monitoring requirements.
- Automated deployment workflows using CI/CD pipelines; slashed release cycle time **30%** and enforced configuration standards; presented deployment best practices to engineering team.

## Projects

### Real-Time Bidding (RTB) System: GBDT CTR Predictors

[GitHub](#)

- Developed production-grade **CTR prediction** models comparing **XGBoost**, **LightGBM**, and **CatBoost** on synthetic RTB data: **XGBoost AUC 0.5662**, **LightGBM LogLoss 0.2237** (best), **CatBoost 0.5494** – all with **sub-0.001ms inference latency** per sample.
- Spearheaded comprehensive **GBDT** benchmark suite measuring AUC, LogLoss, Gini, training time, and **inference latency**; generated production-ready model comparison enabling real-time serving optimization.
- Orchestrated complete RTB engine with second-price auctions, eCPM **ranking**, token-bucket budget pacing, and **<100ms** latency achieving **2000+ QPS** throughput.

### Sponsored Search Ranking: Learning-to-Rank System

[GitHub](#)

- Engineered two-stage **ranking** pipeline: FAISS HNSW candidate retrieval (~3ms) + **TF-Ranking** reranker with **LambdaLoss** for listwise NDCG optimization and IPS position bias correction, improving ranking quality **18%** vs baseline.
- Established PySpark **feature engineering** pipeline with Airflow DAG orchestration for large-scale feature computation; served via FastAPI achieving **<50ms** end-to-end latency on 1M+ queries daily.

### Amazon Product Recommender System: SVD Collaborative Filtering

[GitHub](#)

- Developed large-scale product recommendation system using **SVD-based collaborative filtering** over 1.7M+ user-item interactions, achieving **0.78 NDCG@10** on holdout test set; exposed **inference** via FastAPI with PostgreSQL backend.
- Created Kafka producer/consumer pipeline to stream real-time user interactions and automated model retraining on new batches, reducing stale recommendations **40%**.
- Launched as production-grade serving endpoint with **sub-100ms** recommendation latency, supporting **10K+ daily active users**; articulated system design tradeoffs in technical design review.

### E-Commerce RL Agent: Ranking with Reinforcement Learning

[GitHub](#)

- Optimized RL-based product discovery system using PPO with embeddings, product fetching, and safety filtering; achieved **32% CTR improvement** over collaborative filtering baseline on synthetic e-commerce dataset.

## Education

University of Texas at Arlington – Master of Science in Computer Science

Aug 2023 – May 2025

Andhra University – Bachelor of Technology in Computer Science

Jun 2015 – Apr 2019